Hypothesis tests for two or more means

Overview

Taking stock of where we are and where we are going

Hypothesis test for two means continued

Hypothesis tests for more than two means

Announcement

Homework 7 has been posted!

It is due on Gradescope on Sunday November 8th at 11pm

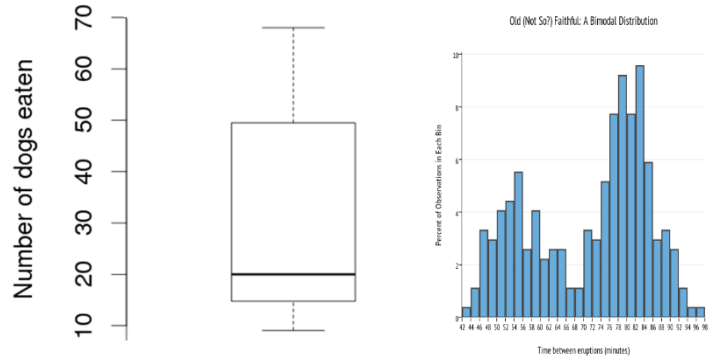
Any questions before we get started…?


WHERE
ARE WE?


Where Are We
Going?

Where we are: what we have covered

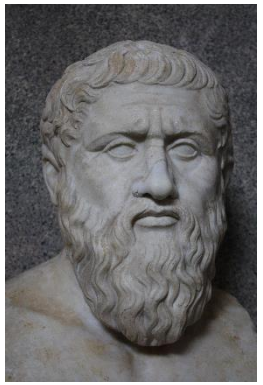
Descriptive statistics

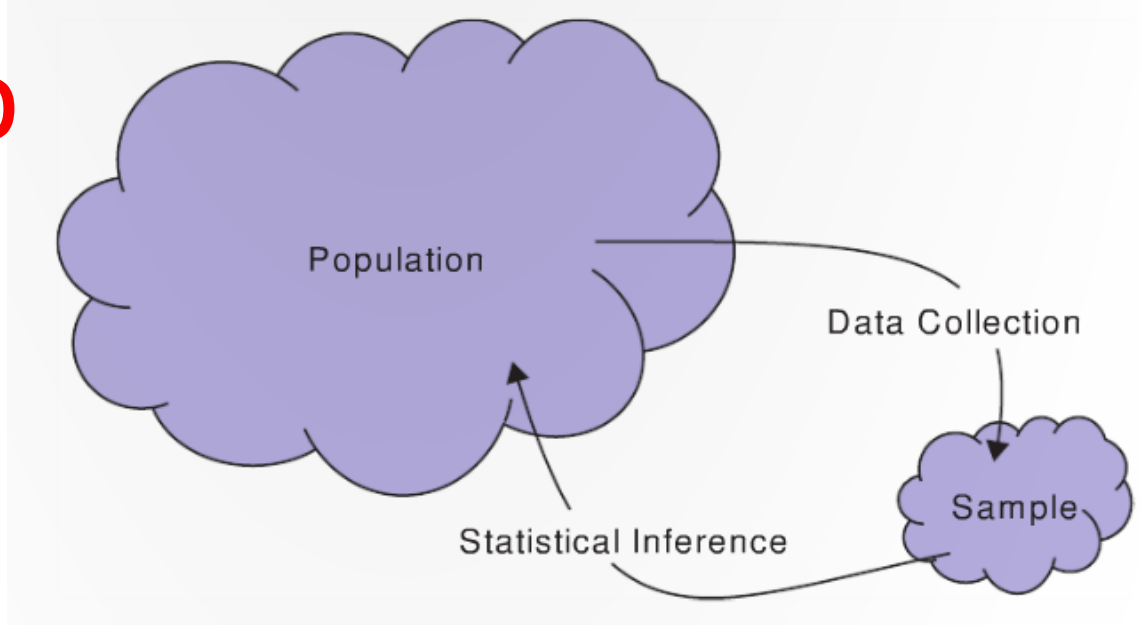


Statistical inference

- We have used computational methods for inference

π, μ, σ, ρ





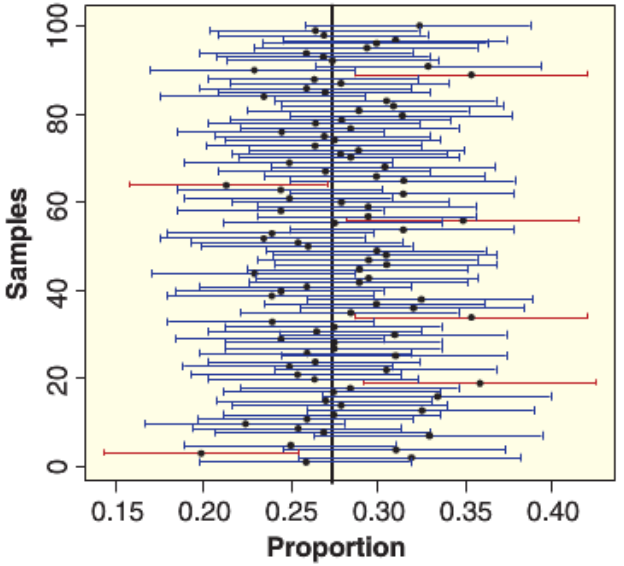
$\hat{p}, \bar{x}, s, r$



Where we are: what we have covered

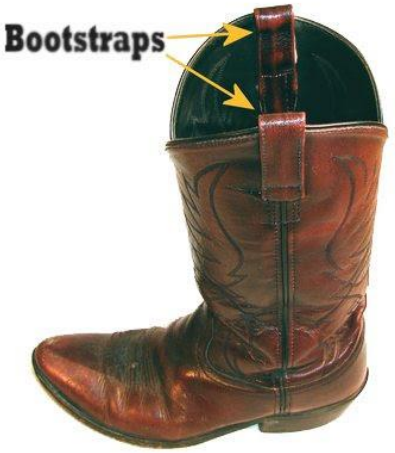
Statistical inference using computational methods

- Confidence intervals using the bootstrap
- Hypothesis tests using permutation/randomization tests
  - Single proportion and two means









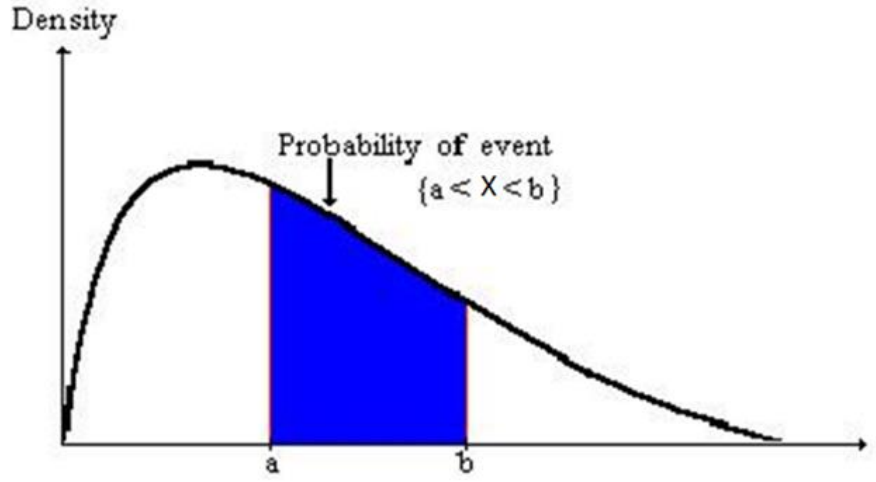
Where we are: where are we going…

Continuation of statistical inference using computational methods

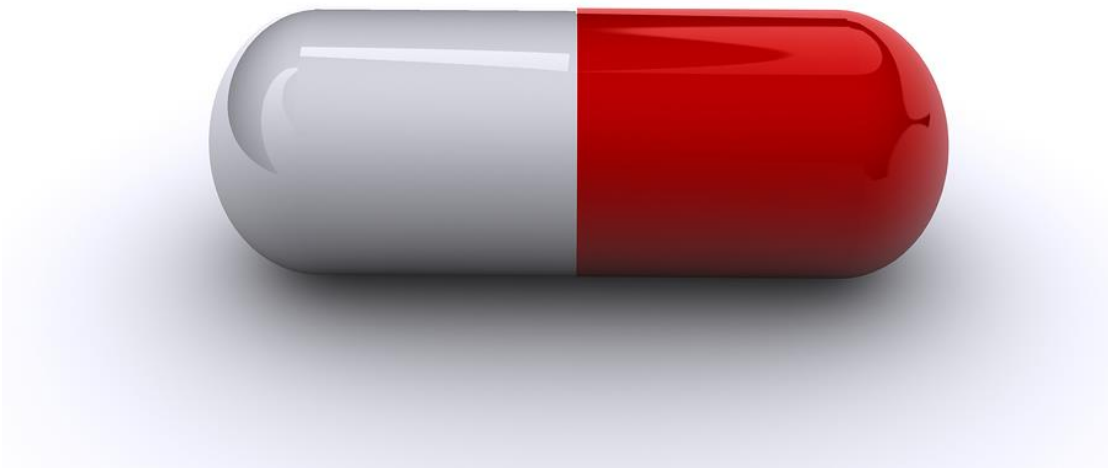
- Hypothesis tests for two or more means and correlation
- Theories of hypothesis tests

Statistical inference based on math/theory

- t-tests, ANOVA, χ^2-test , etc.



Hypothesis tests for comparing two means



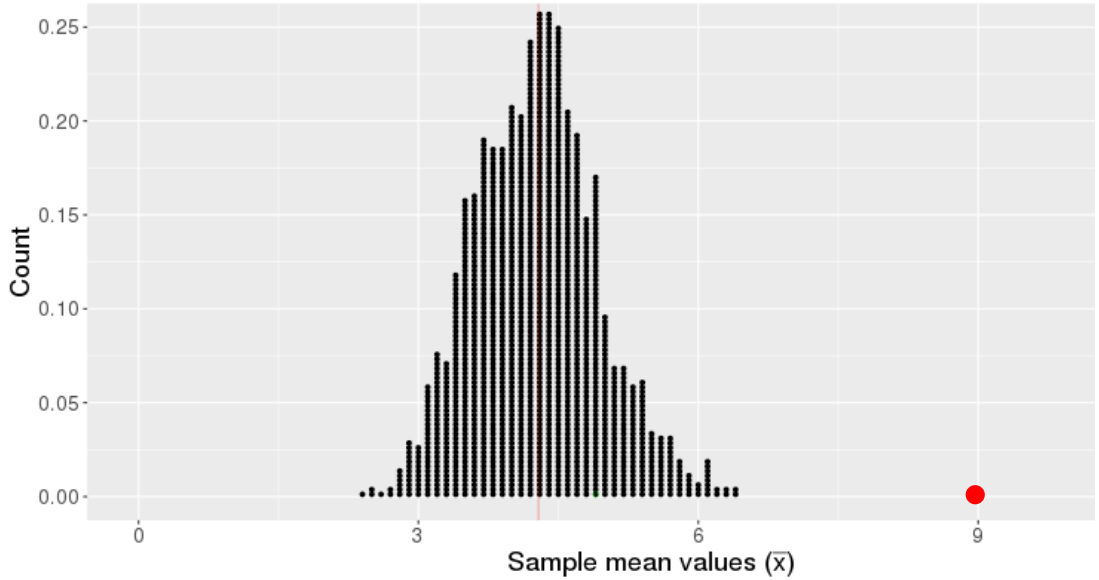
Question: Is this pill effective?

The logic of hypothesis tests…

We start with a claim about a population parameter

- E.g., μ = 4

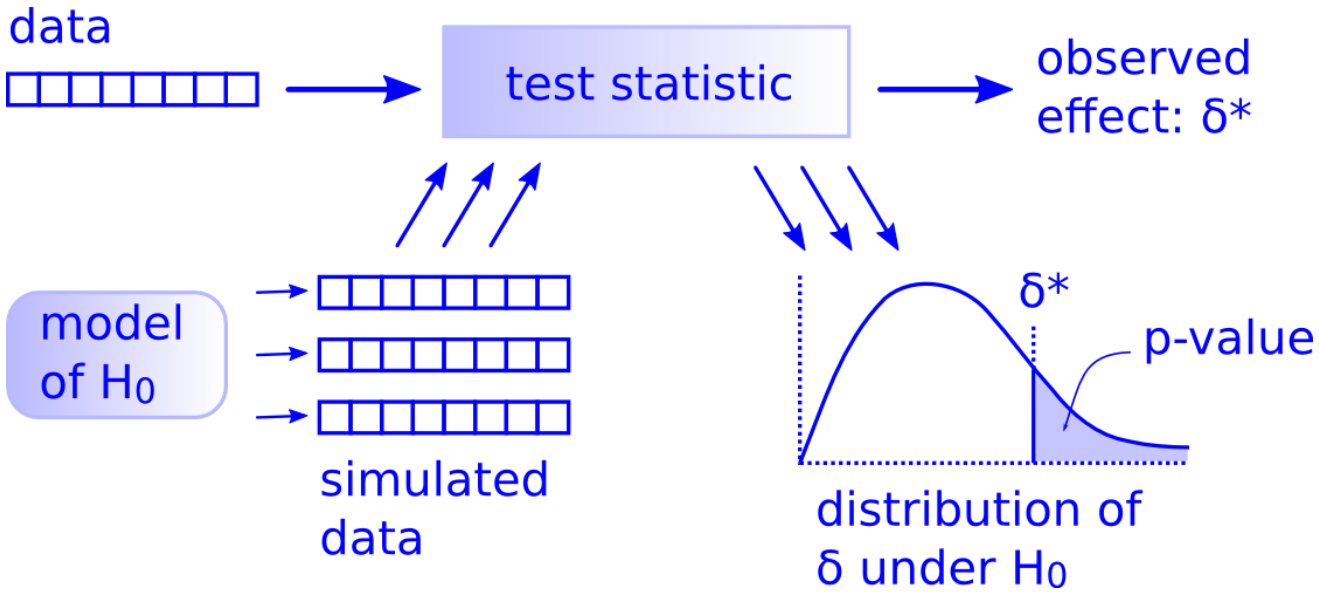
This claim implies we should get a certain distribution of statistics

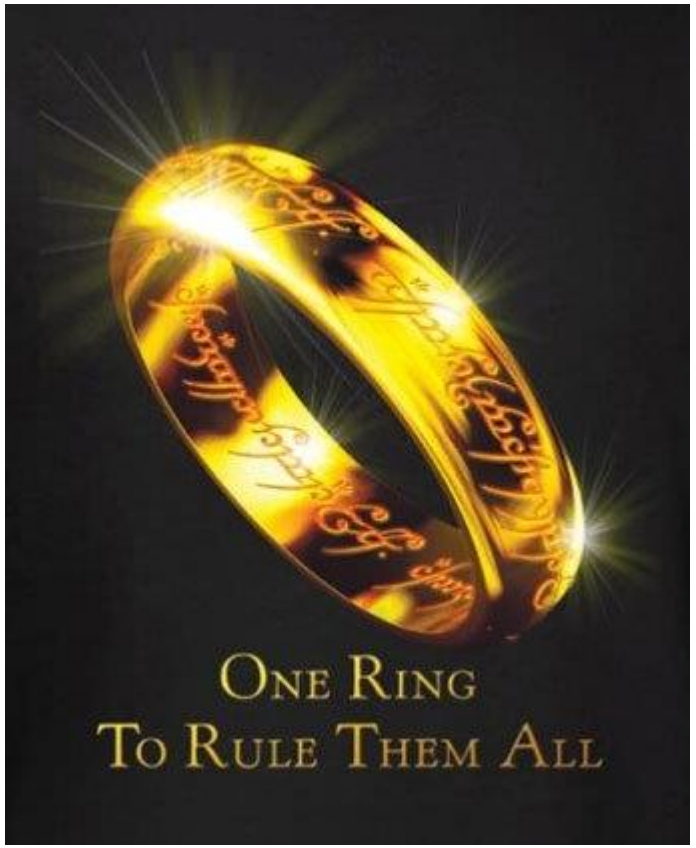


If our observed statistic is highly unlikely, we reject the claim

The logic of hypothesis tests…

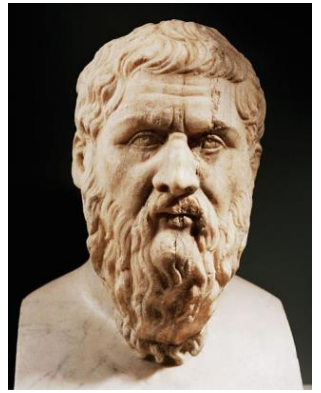
There is only one [hypothesis test](#)!





Just follow the 5 hypothesis testing steps!

Five steps of hypothesis testing

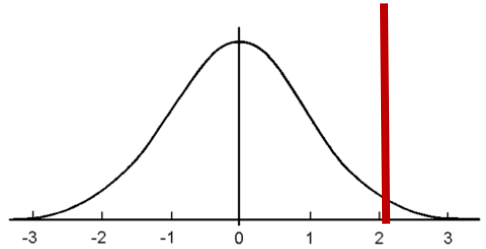


1. State H_0 and H_A
   - Assume Gorgias (H_0) was right



2. Calculate the actual observed statistic

3. Create a **null distribution** of statistics that are consistent with H_0
   - i.e., a distribution of statistics that we would expect if Gorgias is right



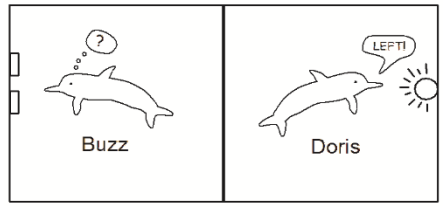
4. Get the probability we would get a statistic more extreme than the observed statistic from the null distribution
   - p-value



5. Make a judgement
   - Assess whether the results are statistically significant

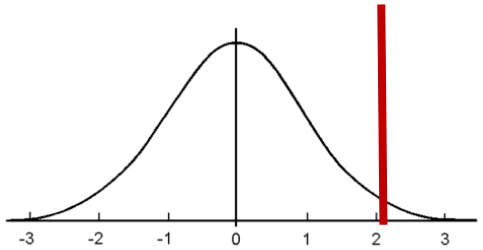
Five steps of hypothesis testing for proportions

1. State the null and the alternative hypotheses
   - H_0: $\pi = 0.5$ (or another value such as 0.6, etc.)
   - H_A: $\pi > 0.5$





2. Calculate the observed statistic
   - $\hat{p} = .937$



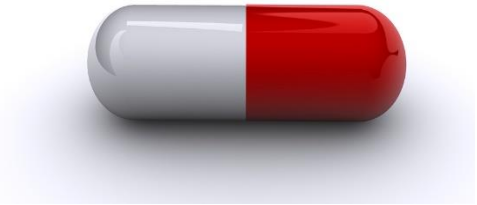
3. Create a null distribution that is consistent with the null hypothesis
   - Null distribution is created by flipping coins: rflip(16, .5)/16

4. Examine how likely the observed statistic is to come from the null distribution
   - pval <- ptial(obs_stat, null_dist, lower.tail = FALSE)

5. Make a judgement
   - If the p-value is small (often less than 0.05) then we can reject the null hypothesis in favor of the alternative hypothesis

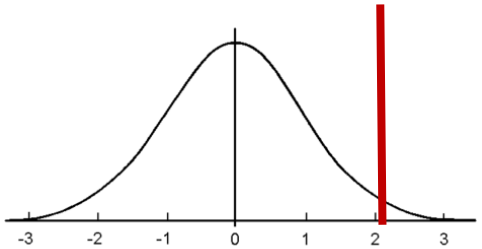
Five steps of hypothesis testing for means

1. State the null and the alternative hypotheses
   - H_0: $\mu_T - \mu_C = 0$
   - H_A: $\mu_T - \mu_C > 0$



2. Calculate the observed statistic
   - $\bar{x}_T - \bar{x}_C$

3. Create a null distribution that is consistent with the null hypothesis
   - Null distribution is created by shuffling (to be explained more soon)



4. Examine how likely the observed statistic is to come from the null distribution
   - pval <- ptail(obs_stat, null_dist, lower.tail = FALSE)

5. Make a judgement
   - If the p-value is small (often less than 0.05) then we can reject the null hypothesis in favor of the alternative hypothesis

Hypothesis tests for comparing two means

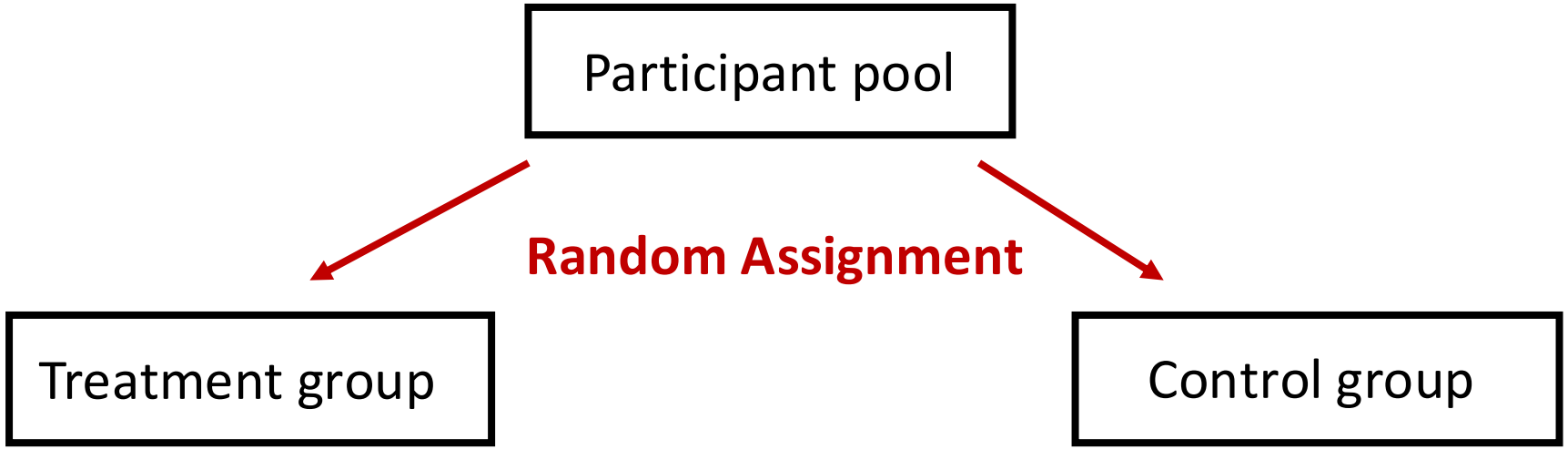


Question: Can we find out the ***Truth*** of whether the pill is effective?

Experimental design: randomized controlled trial

Take a group of participants and ***randomly assign***:

- Half to a *treatment group* where they get and intervention (e.g. a pill)
- Half to a *control group;* e.g., where they get a placebo
- See if there is larger improvement in the treatment group compared to the control group
  - E.g., measure: $\bar{x}_{Effect} = \bar{x}_{Treatment} - \bar{x}_{Control}$

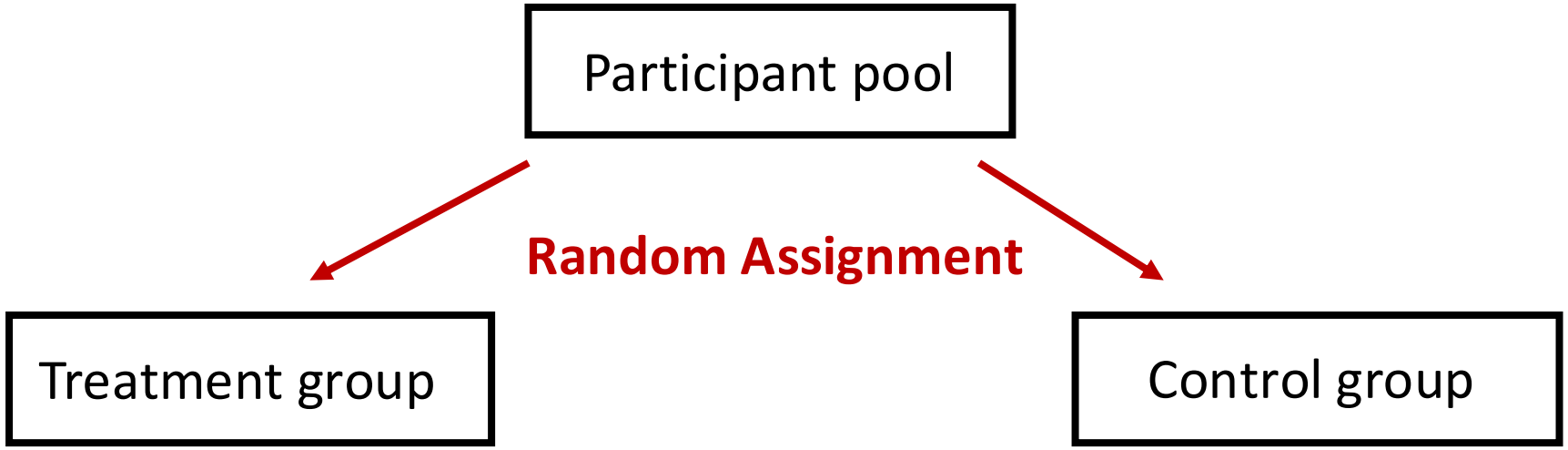


Random assignment allows us to assess causation

Since participants were **randomly assigned** to the treatment and control groups, the only meaningful difference between the groups is our treatment

- E.g., The only (non-random) difference between the treatment and control group is the drug given to the treatment group

Thus, if we used random assignment and there is a statistically significant effect, this allows us to say that the treatment **caused** this observed difference



Example: Does mind-set matter?

A study by Crum and Langer (2007) wanted to assess how one's mindset can affect one's physical health

They recruited 75 housekeepers and randomly assigned them to two groups:

- A treatment group of 41 housekeepers was told that the work they do satisfies the Surgeon General's recommendation for an active lifestyle (which is true)
- The other 34 housekeepers (control group) were told nothing

Let's assess if informing the housekeepers that they were satisfying the Surgeon General's recommendation could **cause** them to lose more weight than the housekeepers who were told nothing

Hypothesis tests for differences in two group means

1. State the null and alternative hypotheses

H_0: $\mu_{Informed} = \mu_{Uninformed}$ or $\mu_{Informed} - \mu_{Uninformed} = 0$

H_A: $\mu_{Informed} > \mu_{Uninformed}$ or $\mu_{Informed} - \mu_{Uninformed} > 0$

2. Write down the statistic of interest using appropriate symbols

$\bar{x}_{Effect} = \bar{x}_{Informed} - \bar{x}_{Uninformed}$

2. Calculate the statistic of interest

After 4 weeks, the exercise habits of the two groups had not changed, but...

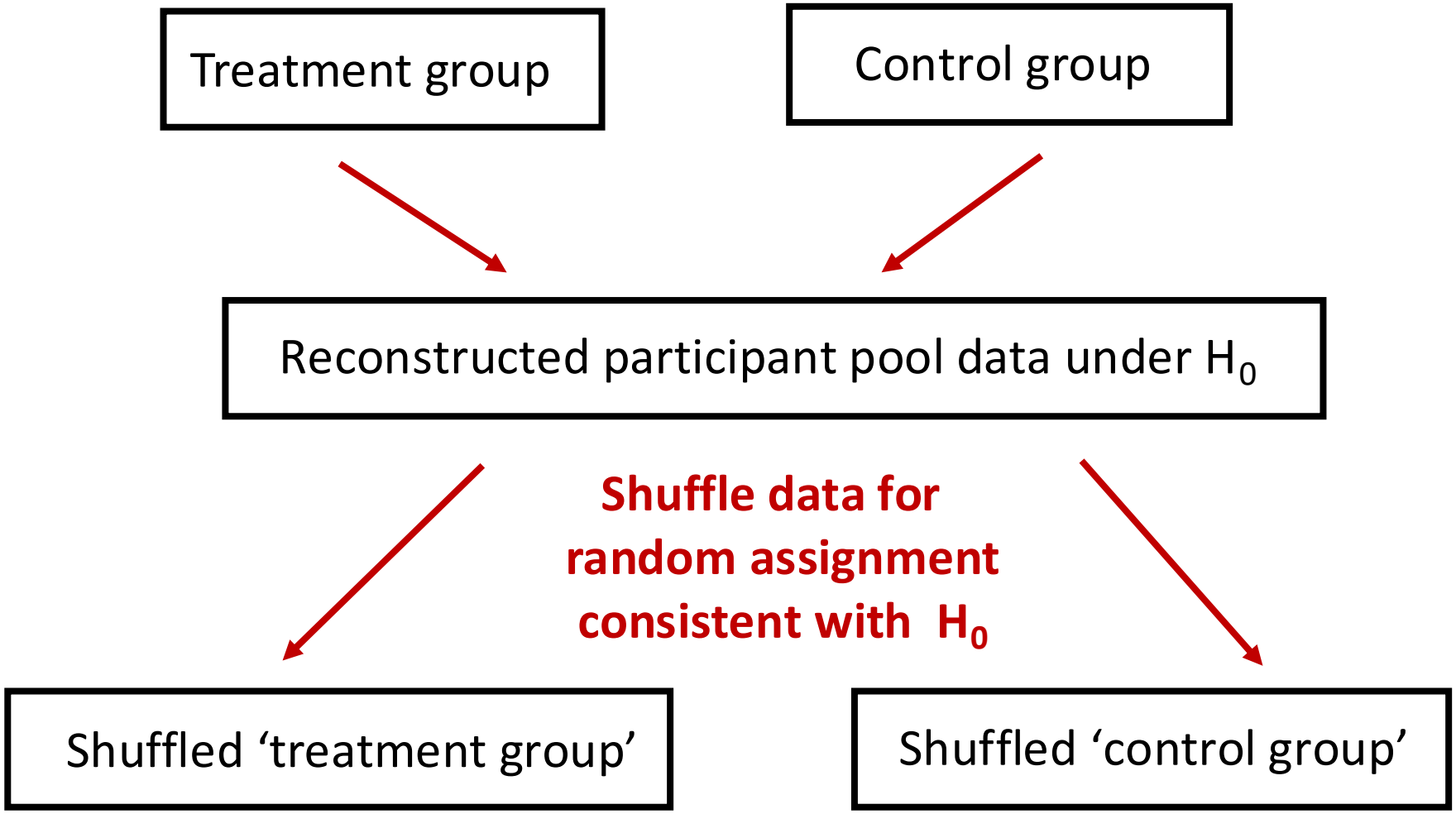
- The informed group on average lost 1.79 lbs (s = 2.88)
- The uninformed group on average lost 0.2 lbs (s = 2.32)

2. What is the observed statistic of interest?

- $\bar{x}_{Effect}$ = 1.79 - 0.2 = 1.59

3. What is step 3?

3. Create the null distribution!



One null distribution statistic: $\bar{x}_{Shuff_Treatment} - \bar{x}_{Shuff_control}$

3. Create a null distribution

1. Combine data from both groups

2. Shuffle data

3. Select the first 41 shuffled points to be the shuffled ‘informed’ group

4. Take the remaining 34 points as the shuffled ‘uninformed’ group

5. Compute the statistic of interest on these shuffled groups

6. Repeat 10,000 times to get a null distribution

3. Creating a null distribution in R

```
# the data from the mindset matters study
load("MindsetMatters_vecs.rda")

informed_weight_loss
uninformed_weight_loss

# observed statistic
obs_stat <- mean(informed_weight_loss) - mean(uninformed_weight_loss)

#  Combine data from both groups
combined_data <- c(informed_weight_loss, uninformed_weight_loss)
```

3. Creating a null distribution in R

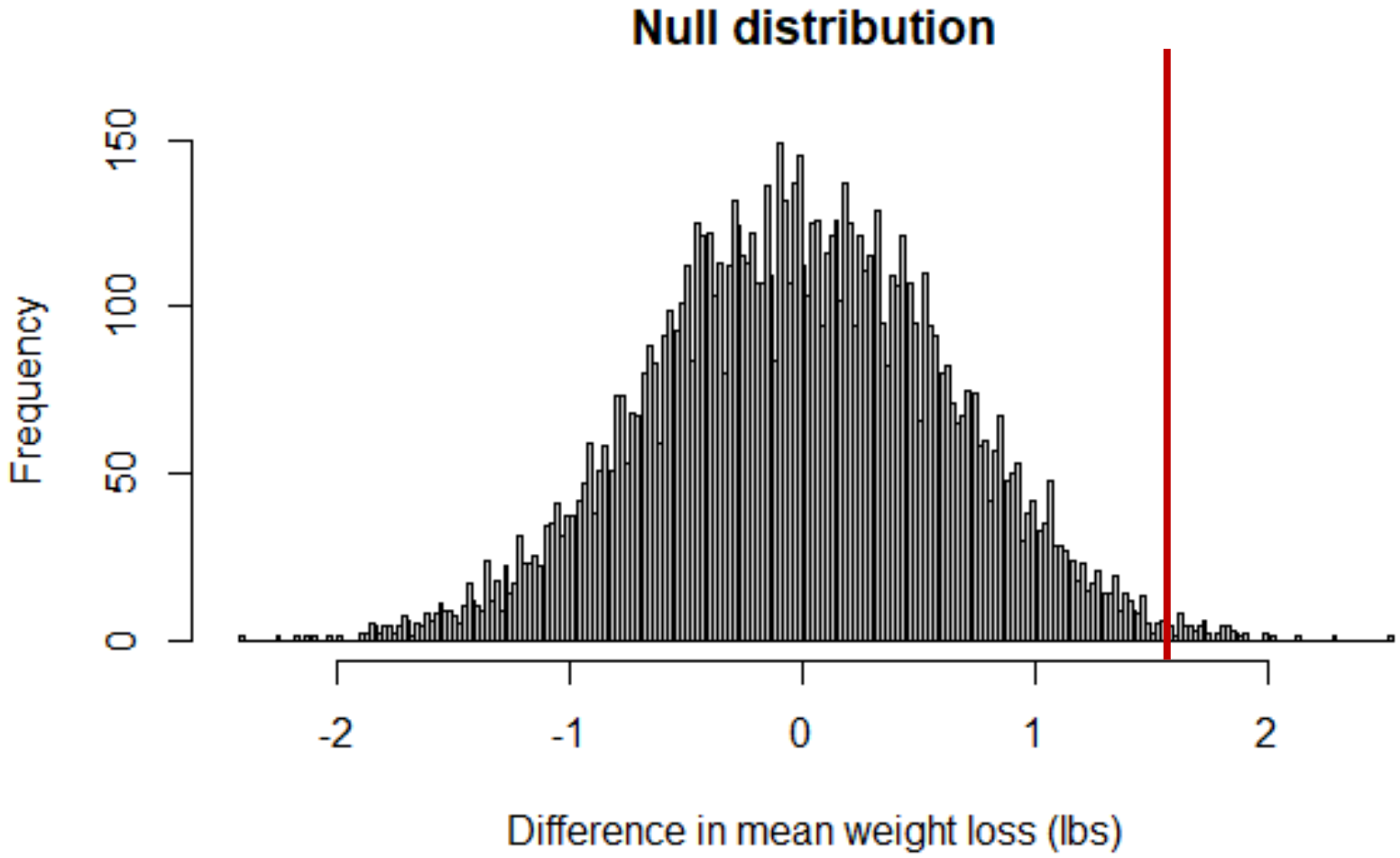
```
null_dist <-  do_it(10000) * {

      # shuffle data
       shuff_data <- shuffle(combined_data)

       # create fake treatment and control groups
       shuff_informed   <-  shuff_data[1:41]
       shuff_control  <-  shuff_data[42:75]

       # save the statistic of interest
       mean(shuff_informed) - mean(shuff_control)

}
```



hist(null_dist, breaks = 200)

Next step?

4. Calculate the p-value

```
# Calculate the p-value
p_value <- ptail(obs_stat, null_dist, lower.tail = FALSE)
```

p-value = 0.005

Next step?

5. Are the results statistically significant?



What can we conclude?

Because **random assignment was used**, we can conclude that informing the housekeepers **caused** them to lose more weight on average

Let’s try it in R…

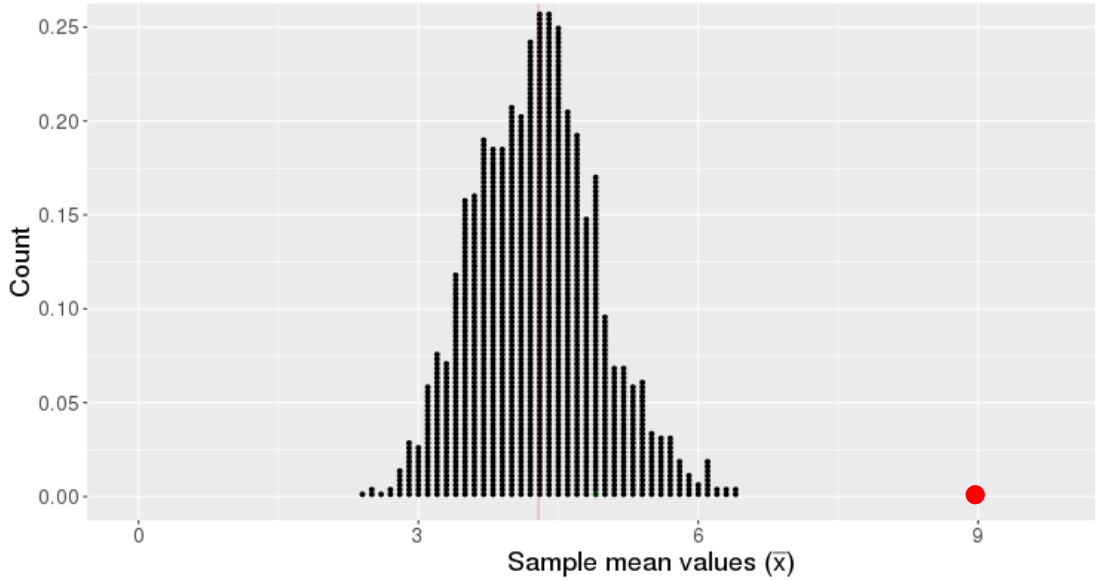
Hypothesis tests for more than two means

The logic of hypothesis tests…

We start with a claim about a population parameter

- E.g., μ = 4

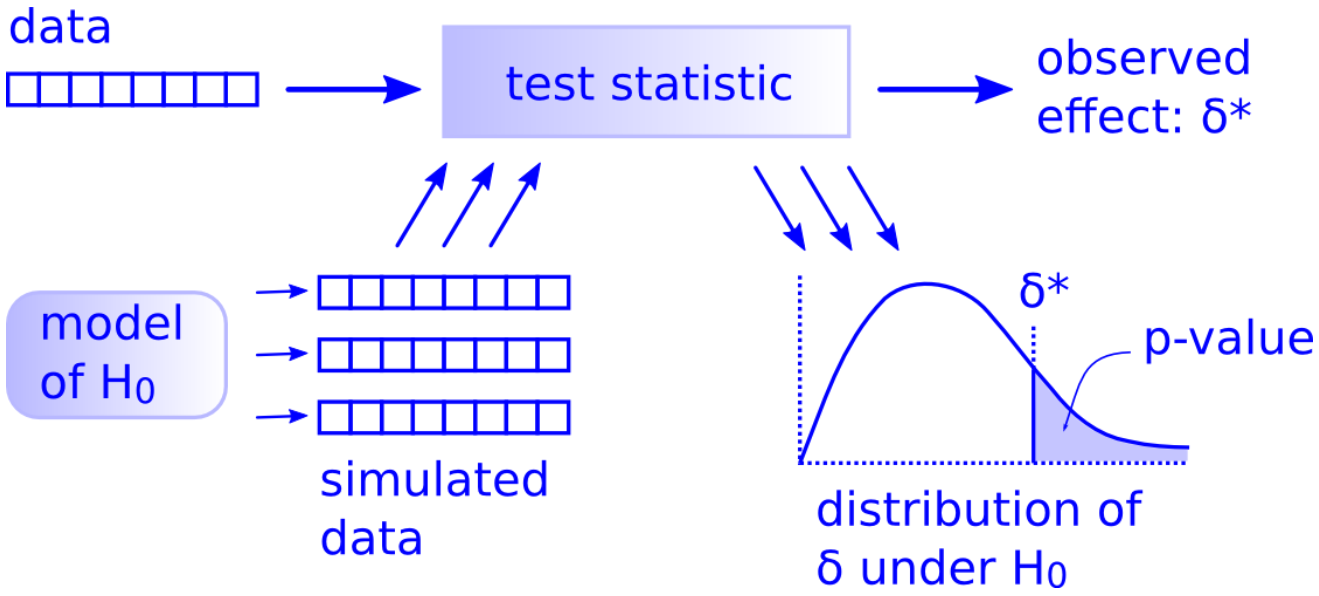
This claim implies we should get a certain distribution of statistics

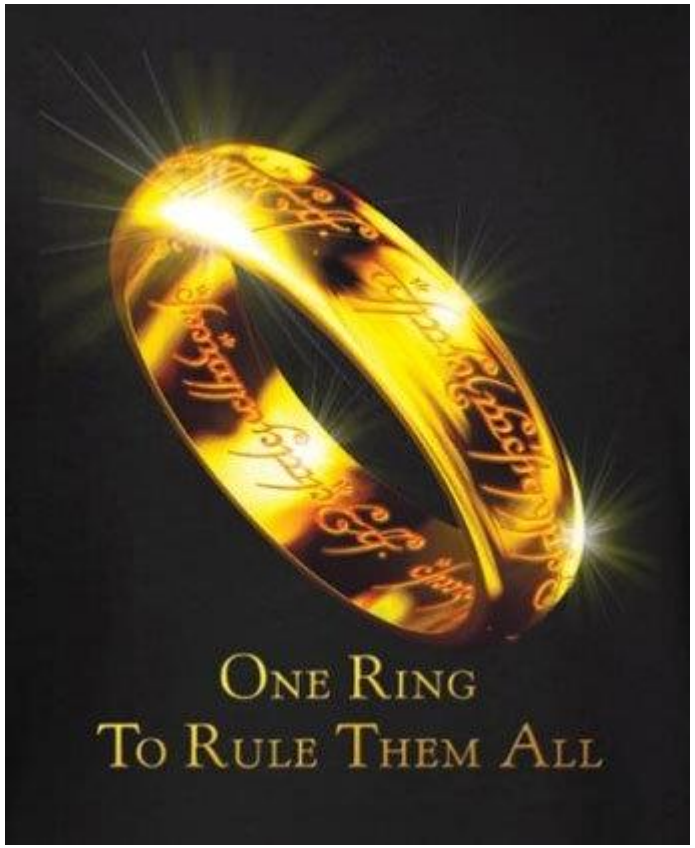


If our observed statistic is highly unlikely, we reject the claim

The logic of hypothesis tests…

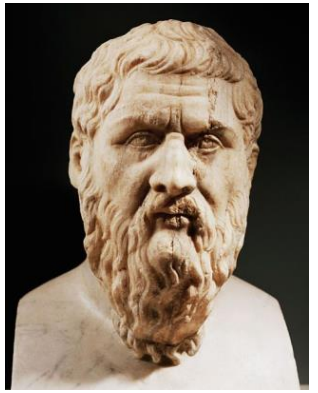
There is only one [hypothesis test](#)!





Just follow the 5 hypothesis testing steps!

Five steps of hypothesis testing

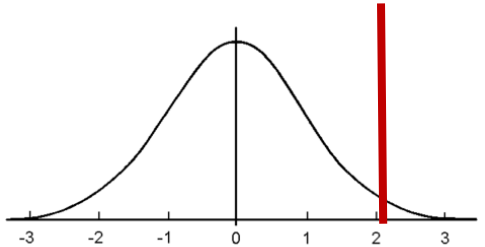


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4. Get the probability we would get a statistic more extreme than the observed statistic from the null distribution
   - p-value



5. Make a judgement
   - Assess whether the results are statistically significant

Comparing more than two means

A group of Hope College students wanted to see if there was an association between a student's major and the time it takes to complete a small Sudoku-like puzzle

	5	3	2		7			8
6		1	5					2
2			9	1	3		5	
7	1	4	6	9	2			
	2						6	
			4	5	1	2	9	7
	6		3	2	5			9
1					6	3		4
8			1		9	6	7	

Comparing more than two means

A group of Hope College students wanted to see if there was an association between a student's major and the time it takes to complete a small Sudoku-like puzzle

They grouped majors into four categories:

- Applied science (as)
- Natural science (ns)
- Social science (ss)
- Arts/humanities (ah)

What is the first step of hypothesis testing?

Sudoku by field

1. State the null and alternative hypotheses!

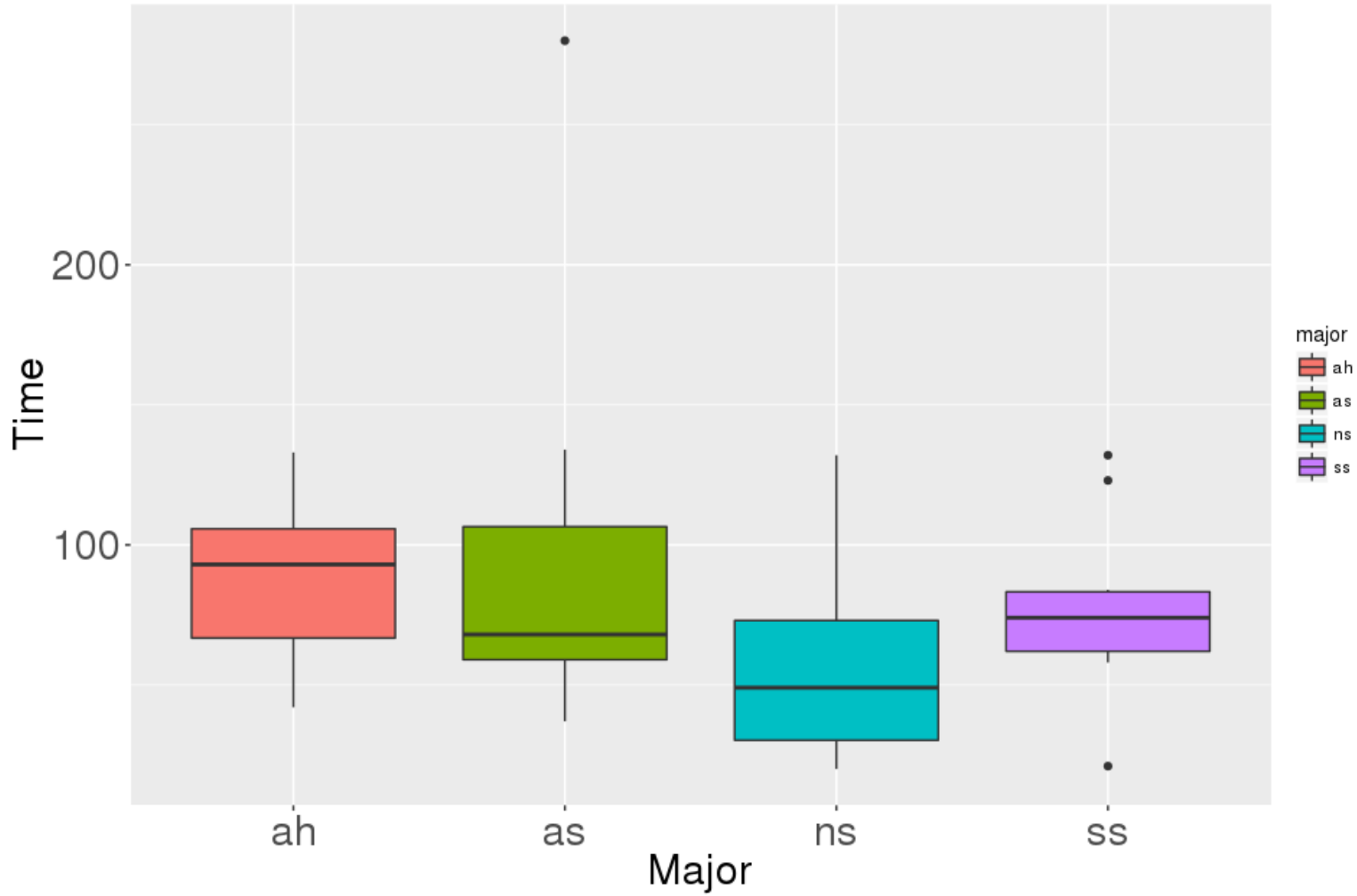
$\mathbf{H_0}$: $\mu_{as} = \mu_{ns} = \mu_{ss} = \mu_{ah}$

$\mathbf{H_A}$: $\mu_i \neq \mu_j$ for at least one pair of fields of study

What should we do next?

Let's plot the data first…

Step 2a: Plot of completion time by major



What should we do next?

boxplot(quant_data ~ cat_data)

Sudoku by field

1. State the null and alternative hypotheses!

$\mathbf{H_0}$: $\mu_{as} = \mu_{ns} = \mu_{ss} = \mu_{ah}$

$\mathbf{H_A}$: $\mu_i \neq \mu_j$ for at least one pair of fields of study

Thoughts on the statistic of interest?

Comparing multiple means

There are many possible statistics we could use. A few choices are:

1. Group range statistic:

   $\max \bar{x} - \min \bar{x}$

2. Mean absolute difference (MAD):

   $(|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$

3. F statistic:

$$F = \frac{\text{between-group variability}}{\text{within-group variability}}$$

Using the MAD statistic

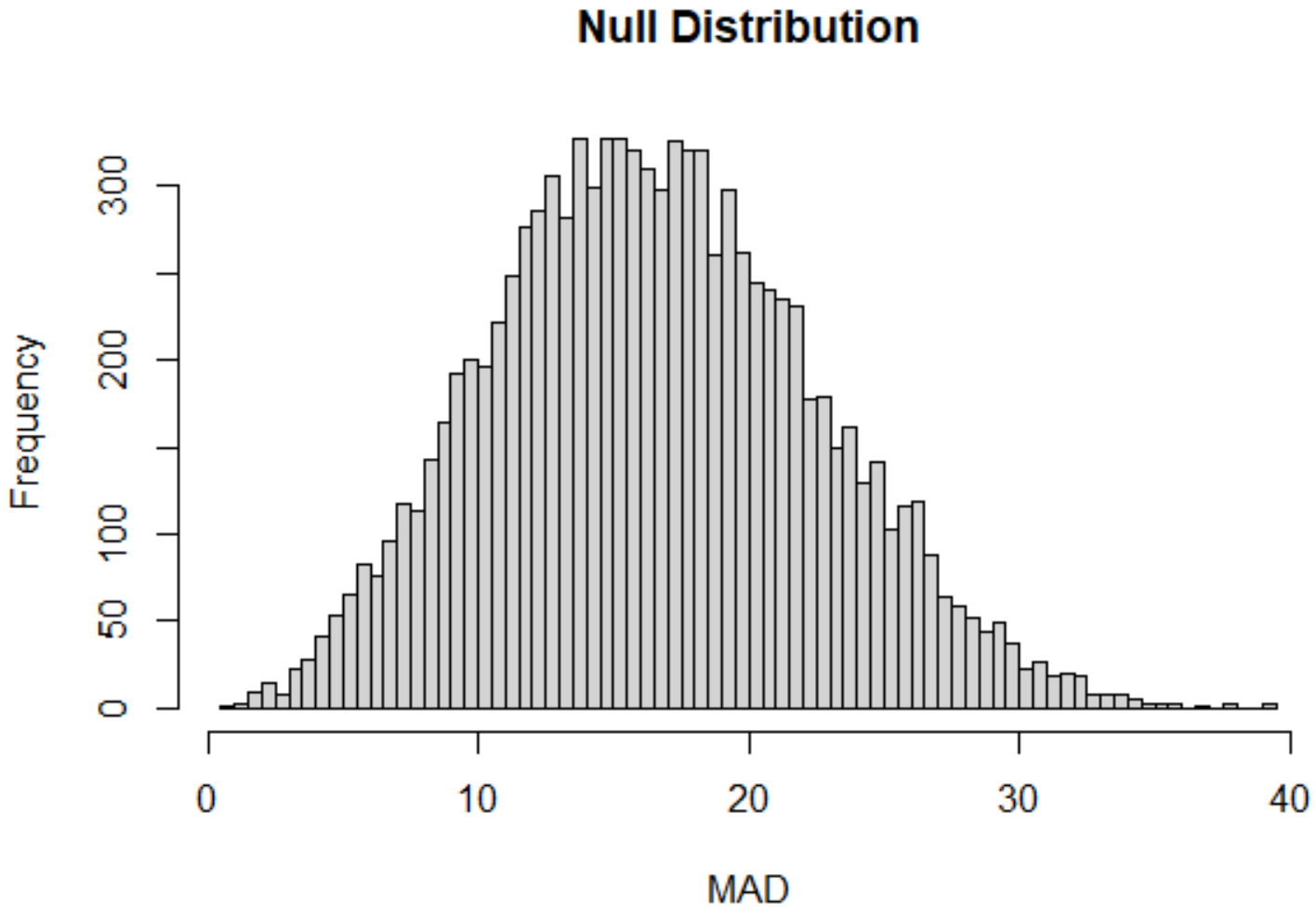
Mean absolute difference (MAD):

$$(|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$$

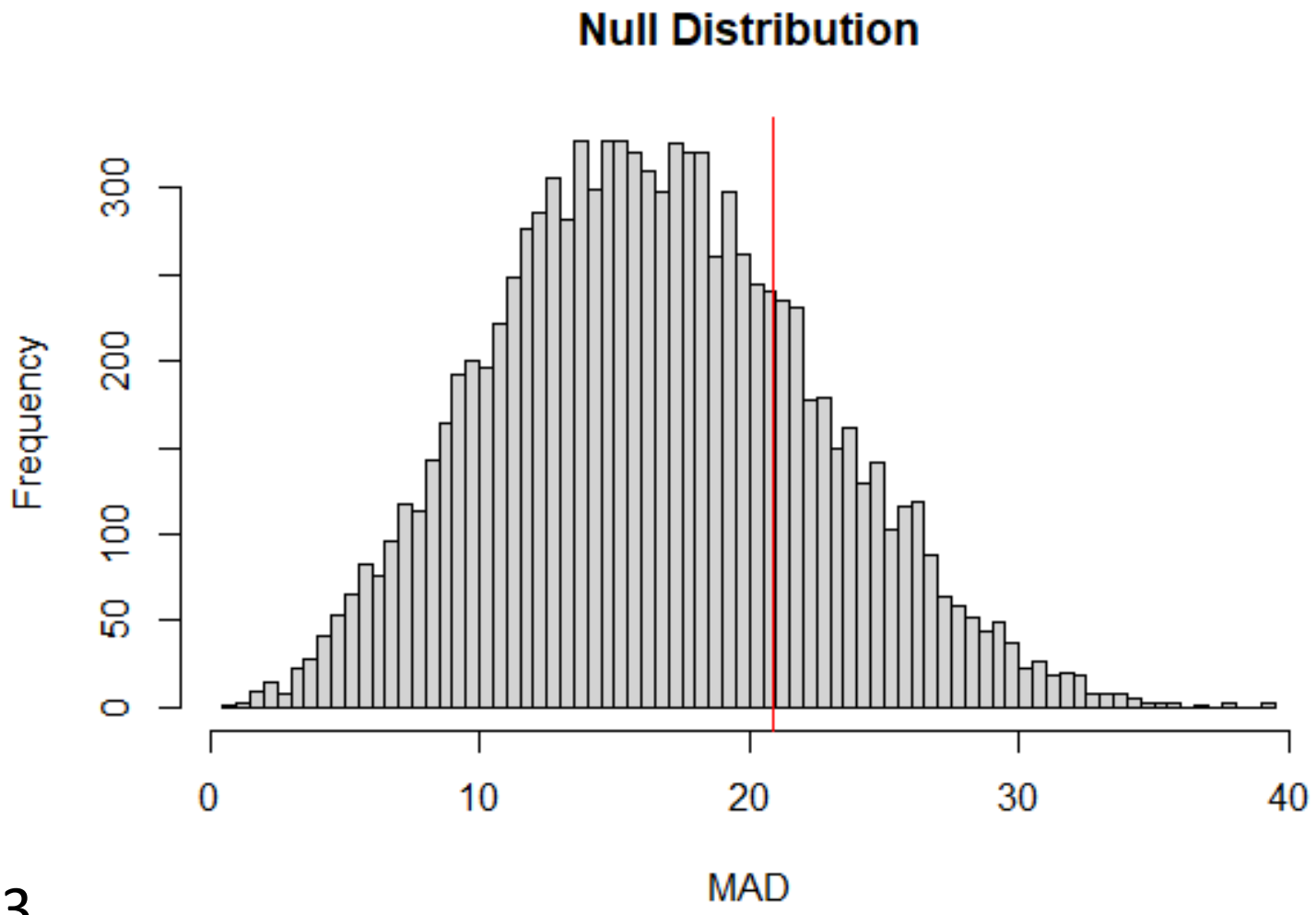
Observed statistic value = 20.88

How can we create the null distribution?

Null distribution



P-value



p-value = .233

Conclusions?



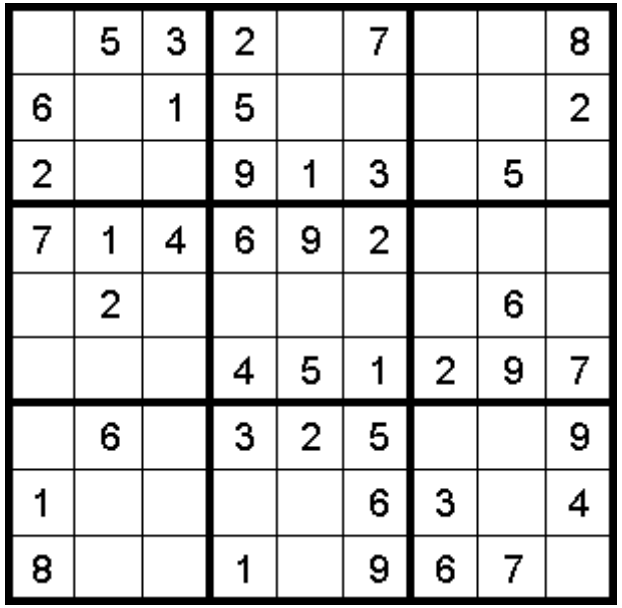
Hypothesis tests for more than two means in R

Hypothesis tests for more than two means in R

Step 1: null and alternative hypotheses…

$\mathbf{H_0}$: $\mu_{as} = \mu_{ns} = \mu_{ss} = \mu_{ah}$

$\mathbf{H_A}$: $\mu_i \neq \mu_j$ for at least one pair of fields of study



Let’s try this analysis in R…

```
# get the data
sudoku_data <- read.table("MajorPuzzle.txt", header = TRUE)

# Extract vectors from the data frame
completion_times <- sudoku_data$time
majors <- sudoku_data$major
```

major	time
ss	21
as	280
ah	96
as	87
ah	133

Visualize the data

How can we visualize the data?

```
# We can create side-by-side boxplots using…
boxplot(completion_times ~ majors,
        xlab = "Major",   ylab = "Time (s)")
```

major	time
ss	21
as	280
ah	96
as	87
ah	133

Calculating the statistic of interest

We can get the MAD statistic using the get_MAD_stat() function

$$MAD = (|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$$

get_MAD_stat(data_vector, grouping_vector)

- data_vector: a vector of quantitative data
- grouping_vector: a vector of categorical data indicating which group the quantitative data is in

major	time
ss	21
as	280
ah	96
as	87
ah	133

Can you get the MAD statistic for the sudoku data?

```
obs_stat <-  get_MAD_stat(completion_times, majors)
```

Creating the null distribution

Q: How could we create one point in a null distribution?

- A: Shuffle the grouping_vector (major vector) and calculate the MAD statistic

Q: How can we do this in R?

```
shuffled_majors <- shuffle(majors)

get_MAD_stat(completion_times, shuffled_majors)
```

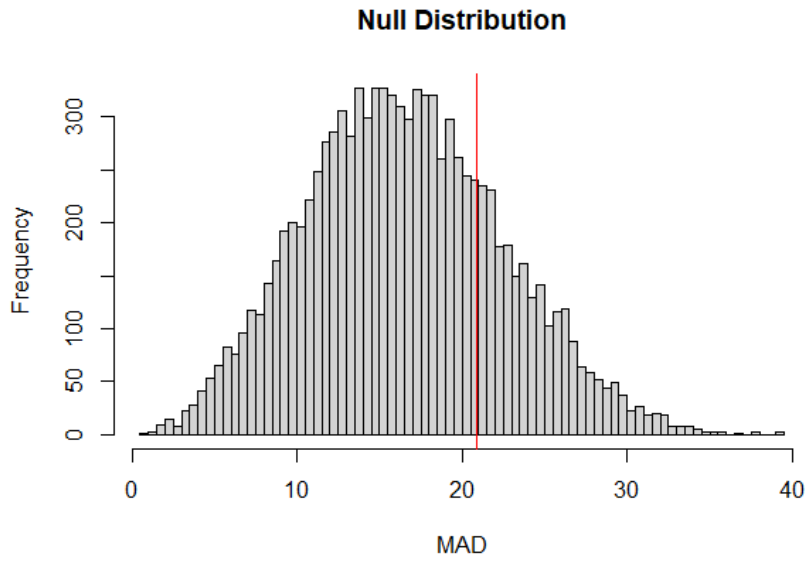
major	time
ss	21
as	280
ah	96
as	87
ah	133

Creating the null distribution

Q: How can we create a full null distribution?

```
null_dist <- do_it(10000) * {
    shuffled_majors <- shuffle(majors)
    get_MAD_stat(completion_times, shuffled_majors)
}

# visualize the null distribution
hist(null_dist, breaks = 200)
abline(v = obs_stat, col = "red")
```



Steps 4 and 5

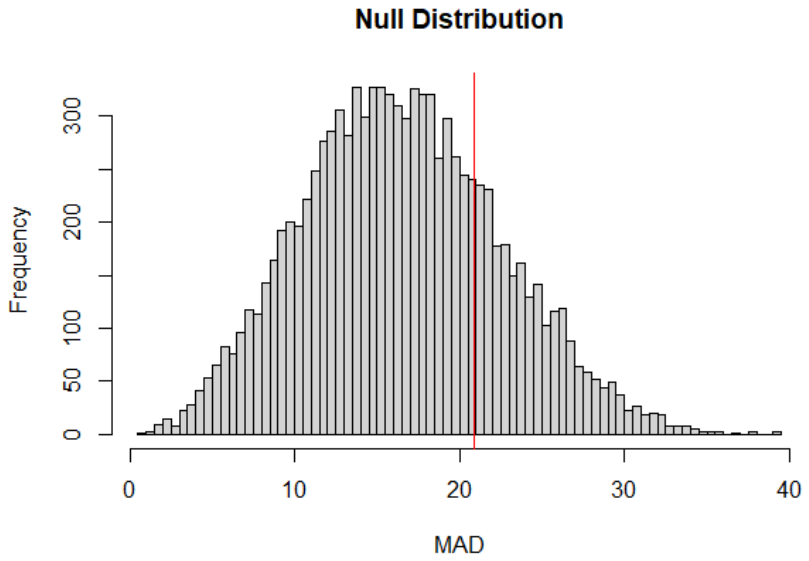
Q: What do we do next and how do we do it?

- A: We get the p-value

ptail(obs_stat, null_dist, lower.tail = FALSE)

$$MAD = (|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$$





Let's try it in R!